

DRAFT SUPPLEMENT B (For Schematics, Design Development, Construction Documents, Construction Cost Estimate and Construction Period Services)

DESIGN TIMEFRAME: A/E shall deliver final construction documents **No Later Than (NLT) 365 days** after the Award/Notice to Proceed Letter.

CONSTRUCTION BUDGET/DESIGN LIMITATION: **\$ 5,000,000- \$10,000,000.** The A/E will also be required to incorporate deductive alternate bid items into the construction documents that will allow for up to a 20% cost reduction.

B1 SCHEDULE OF SERVICES:

I. INTRODUCTION:

The mission of the Veterans Affairs Medical Center (VAMC) in Marion, Illinois is to provide the highest quality health care to our nation's veterans. In order to accomplish this mission, Marion VAMC Engineering Service intends to partner with an A/E Firm to relocate Cardiac Cath Lab from Building 1 second floor to building one fourth floor at the Marion VAMC Campus, 2401 West Main Street, Marion, IL 62959.

The A/E Firm must provide Basic Professional Services as delineated in the Scope of Work. As part of the Basic Professional Services, Marion VAMC Engineering Service expects the following primary A/E disciplines to be included: Supervisory, VPIH, Civil/Structural, Architectural, Mechanical, Plumbing and Electrical, Physical Security and Fire Protection.

II. SCOPE OF CONSTRUCTION PROJECT:

1. **Existing Conditions:** Fourth floor Building 1 currently accommodates (3) operating rooms, an Endoscopy procedure room and recovery room. The ward also contains patient care support spaces, including: Reception room, Recovery room, EMS closets, locker rooms, linen closet, nurse station, nursing staff administrative space, nourishment, local pharmaceutical dispensary, and utility spaces. The floor area of 4th floor is approximately 5,320 gross square feet (gsf).
2. **Project Intent:**
 - a. **Survey/Site Investigation:**
 - i. A survey/investigation shall be undertaken to fully assess site conditions and existing conditions of the proposed project area.
 - ii. Investigation into other potential system upgrades will occur concurrently. Upgrades could be driven by advances in technology, safety, energy savings or any other factors which would improve overall system performance within the limitations of the project intent.
 - iii. Investigation shall include documenting interior station utilities that may be affected during construction activity.
 - b. **General Conditions:**

- i. The A/E shall investigate the impact of infection control and maintaining hospital mitigation of air quality requirements, vibration and noise pollution.
 - ii. The contractor will be responsible for field verification of existing utilities related to this project's requirements. If any utilities are damaged during construction activity all mitigative repairs shall be the responsibility of the contractor.
 - iii. VA Engineering Service personnel will assist when site inspection procedures affect the operations of any of the building utility systems.
 - iv. All utilities are critical to facility operations. Any investigation or work impacting the facility or requiring a shutdown shall be scheduled **twenty one (21)** days in advance with written notice. Work shall be performed at nights and on weekends. The COR has the right to stop work after 8 hours of planned medical facility outages.
 - v. All proposed rework as well as the final, as-built conditions of the building shall be subject to comment, revision and approval by the Illinois State Historic Preservation Agency. All costs and labor incurred to achieve a design approved by the Historic Preservation Agency shall be the responsibility of the A/E.
 - vi. Design errors or omissions, which result in damages or additional cost to the Government will be evaluated for potential A/E liability. If the Government determines that the A/E is financially liable for a design deficiency, the Government will advise the A/E by official correspondence. The Department of Veterans Affairs actively pursues reimbursement of costs incurred by the Government as result of the A/E's errors and/or negligent performance. The preferred method of settlement of A/E financial liability is for the A/E to negotiate directly with the Contractor and provide the result to the Department of Veterans Affairs for review and approval. Where the A/E cannot reach an agreement with the Construction Contractor or if the A/E declines to negotiate, The Department of Veterans Affairs will arrange settlement directly with the Contractor and will bill the A/E.
 - vii. All equipment and finishes to be included in the project design shall be in accordance with current Marion VAMC standards and existing maintenance contracts.
 - viii. The normal operating hours of this facility are 7 am to 4:30 pm, Monday through Friday, excluding recognized holidays. The construction documents shall require the contractor to reflect this in his proposed work schedule. Work outside the normal operating hours shall require prior authorization by the VAMC.
- c. Demolition:
- i. As determined during design, all existing materials removed during construction will be disposed of according to the direction of the construction documents and as directed by VA staff.
- d. Industrial Hygiene:
- i. Provide an abatement plan for any hazardous materials (asbestos, lead, etc.) which are known prior to or discovered during the investigation and construction phases.
- e. Structural:
- i. Determine capacity of existing building structure to accept the upgraded mechanical components proposed, if needed.

- ii. If weaknesses or inabilities to accept the components are encountered, modifications to the structure to allow for the upgraded component would likewise be proposed as part of the project documents.
 - iii. Determine anchorage and bracing necessary to resist seismic loads for all material and components added by this project in accordance with the VA Structural Design Manual (PG-18-10).
- f. Architectural:
 - i. The scope of architectural work shall include all work architectural work necessary to move Cath Lab and effectively update the spaces included in the project area.
- g. Mechanical:
 - i. The scope of mechanical work shall include updates to all existing mechanical systems that serve the proposed project area.
 - ii. HVAC system capable of meeting most recent VA HVAC design requirements for 100% outside air in patient care spaces.
- h. Plumbing:
 - i. The scope of plumbing work shall include updates to all plumbing systems, fixtures, outlets, disposal and ventilation systems, including:
 - i. Domestic water systems – piping, fixtures, etc.
 - ii. Medical gas systems – including piping, valves, outlets, panels, etc.
 - iii. Fire Protection systems
 - iv. Waste water disposal/sanitary sewer
- i. Electrical/IT/Telecom:
 - i. The scope of electrical/IT/telecom work shall include updates to all existing electrical/IT/telecom systems that serve the proposed project area.
 - ii. Provide construction period services for arc flash and protective device coordination for all equipment under the project scope. After new equipment selection, the A/E shall provide the aforementioned services in addition to typical design phase site investigation and construction phase site visits. Submit TCC's to the COR for approval. Refer to the VA Electrical Design Manual (PG-18-10) section 1.10. Include VA Master Spec (PG-18-1) 26 05 73.
- j. Construction Phasing:
 - i. Construction contract will utilize critical path phasing as necessary to minimize interruption and disturbance to the building and its occupants.
 - ii. Phasing shall be determined through collaborative effort of both the A/E firm and the VAMC Engineering Service.

III. SCOPE OF WORK FOR A/E:

1. Design Development:

- a. The A/E shall make investigations necessary to thoroughly evaluate the area. All designs shall be in accordance with A/E Submission Instructions for Minor and NRM Construction Program, Program Guide PG-18-15, Volume C.
- b. Upon completion of initial field survey and assessment, provide a comprehensive report of existing conditions and findings to Marion VAMC Engineering Service. Report will include initial cost estimate for all work included in the **Relocate Cath Lab to Building 1 Fourth Floor**. This report will be considered the Design Narrative as referenced in "B5 DESIGN REVIEW AND COMPLETION SCHEDULE" below.
- c. Design shall comply with the most current version of the VA Physical Security & Resiliency Design Manual at time of award found on the VA technical Information Library:
<https://www.cfm.va.gov/til/dManual.asp>
- d. General Statement: The A/E shall prepare plans and specifications for a construction contract to accomplish the following work:
 - I. Renovate 4th Floor Building to comply with the latest patient care standards and requirements
 - II. Provide a design to include one operating room, and 5 Prep/Recovery Bays.
 - III. Provide a design to accommodate and update all existing patient and staff support spaces required to comply with facility needs and VA requirements for a functional Cath Lab. An approximation of the program needs has been listed in the Table below.

Cath Lab - Program for Design

Cath Lab	Quantity	NSF	Program for Design
Anesthesia Equipment Room	1	120	120
Anesthesia Team Room	1	120	120
Break Room	1	120	120
Clean Linen Room	1	80	80
Clean Utility Room	1	80	80
Consult/Teaching	1	120	120
Control Room (with PACS Viewing Workstation)	1	120	120
Crash Care Alcove	1	20	20
Equipment Storage	1	120	120
Family Restroom	1	80	80
Female Locker Room	1	100	100
Female Restroom	1	60	60
HAC	1	60	100

Handwashing Station (Scrub Room)	1	10	80
Imaging Equipment Alcove	1	30	30
Logistic Storage	1	120	120
Male Locker Room	1	100	100
Male Restroom	1	60	60
Med Equipment Room	1	120	120
Nourishment Room	1	60	60
Operating Room	1	750	750
OR Equipment Room	1	120	120
Patient Team Room (Patient Procedure Room)	1	120	160
Prep/Recovery Bays	4	120	480
Service Chief	1	100	100
Provider Workroom	1	120	120
Reception	1	85	85
Refreshment Alcove	1	40	40
Sterile Storage	1	100	100
Patient Restroom	1	60	60
Soiled Utility	1	80	80
Waiting Room	1	160	160
Wheelchair/Stretcher Alcove	1	30	30
Totals			4095

4th Floor Available Sq Ft	5430		
Reserved Corridor Sq Ft (154'x8')	1232		Difference
Program for Design Available Sq Ft	4198		103

- IV. Provide a design to update all systems within Fourth floor building 1, including but not limited to:
 - i. Mechanical, Electrical, Plumbing (MEP)
 - ii. Architectural
 - iii. Information Technology (IT)
 - iv. Fire Protection
 - v. Physical Security
- V. Provide a design that accommodates all the latest VA standards and code requirements for spaces intended for standard and intensive inpatient care.
- VI. Design the Government's Guiding Principles for Sustainable Federal Buildings standards with verification by a Professional Engineering stamp and completion of the Guiding Principles Checklist. Architect is responsible for verifying that the construction meets the Guiding Principles for Sustainable Federal Buildings.
- VII. Full commissioning per the VA Whole Building Commissioning Process Manual as applicable to this building is required. All associated costs and labor (Commissioning

agent(s), inspections/witnessing, and documentation) shall be the responsibility of the Architect. Provide cost estimates associated with deduct alternates.

- VIII. Design multiple deductive bid alternates into the construction documents equal to at least 20% of the total construction cost. The base design must be fully functional with or without the alternates. Provide an itemized cost estimate associated with each deduct alternate.

2. Construction Documents:

- a. After the VAMC approves the A/E's conceptual design, the A/E shall prepare construction documents necessary to accomplish the approved conceptual design. These documents shall bear the seals and signatures of all licensed individuals responsible for the content of the construction documents. Application of the seals and signatures shall be considered verification that the design was conducted in accordance with all applicable code requirements, standards and best engineering practices.
- b. As part of the 100% design submission, the A/E shall provide a written synopsis describing the overall scope of the construction project as well as any pertinent information about the project that would assist potential bidders in preparing a comprehensive bid package. This synopsis is intended to be a general description of the project not to exceed two written pages in length.
- c. The final design documents from a resultant task order will be subject to review and approval (by signature) of all members of an Integrated Project Team (IPT) consisting of medical center personnel. These individuals will be identified by the COR at the first design review meeting.

3. Construction Period Service: The A/E shall perform construction period services as detailed later in this scope of work.

4. Functional Product:

- a. It is important to emphasize that the A/E shall perform design development and detailed design work, and prepare detailed construction documents, keeping in mind that the end result is a completely functional and fully operational system.
- b. VPIH Survey: The A/E Firm will complete a survey, to include testing, identifying any and all asbestos and lead containing materials within the project area. This project will remove all lead and asbestos in areas being renovated.
- c. All spaces within the scope of work must be designed according to applicable codes and standards.
- d. The A/E shall submit a design narrative (15%) and design plans (30%, 60%, 90%, 100% Contract Docs for Bidding, and Release for Construction Set) specifications, submittals schedule, estimates and other documentation in accordance with the "B4 DESIGN DELIVERABLES" section of this Supplement B. The construction work will be performed under a separate Firm-Fixed-Price contract awarded by the VA. As a result, the Construction Documents must conform to the estimated Total Construction Cost Price, not including any

contingencies, and convey the entire construction work completely and clearly, such that any bidder can produce an accurate bid without surveying the work site.

- e. The A/E will also participate in the Marion VAMC Pre-Construction Risk Assessment to determine the requirements for Infection Control, Interim Life Safety Measures, occupational safety and health, patient safety, public safety, security controls for physical and logistical access, and environmental controls and limitations on vibrations, noise, and disruptions to operations and utilities. The results of these assessments will be included in documentation submitted by the A/E pursuant to, and inclusive of, the Construction Documents.
 - f. Prior to submittal of a response to this proposal the A/E shall meet with the Contracting Officer's Representative (COR) at the proposed project site to review the scope of services and the project goals. The COR will provide A/E with a copy of the portions of the latest Facilities Condition Assessment (FCA) applicable to this project.
 - g. Overall site plans will be provided as a Reference Drawing with Buildings showing the areas within the project boundary.
 - h. The C&A requirements do not apply, and that a Security Accreditation Package is not required.
5. Sustainability Requirements:
- a. Design shall comply with the most current version of the VA Sustainable Design Manual at time of award found on the VA technical Information Library
<https://www.cfm.va.gov/til/sustain.asp>
 - b. The following deviations are allowed: Not applicable.
 - c. GUIDING PRINCIPLES CHECKLIST: A/E shall complete and submit the appropriate Federal Energy Management Guiding Principles Checklist:
<https://energy.gov/eere/femp/downloads/guiding-principles-checklists-new-construction-and-modernization-and-existing>

IV. A/E FEE DETERMINATION:

The A/E is to present their fee proposal for evaluation and negotiation in the following format:

1. Submit completed Price Proposal on NCO 15 Form 10-6298. The spreadsheet is to have the work broken out in hours by position, by discipline and by design phase. Include a list of the number of drawings required for each design phase broken out by drawing type per discipline. Include a summary of the foregoing with the number of hours required for each drawing type and each major task. The spreadsheet will be used to facilitate review and comprehension of work effort and document fee. Include Part IV in cover letter and Part V (signed) as a separate attachment. Submit the electronic file of the spreadsheet upon request.

2. Include a letter of understanding with the price proposal that presents a detailed understanding of the Supplement B. The letter of understanding shall include an analysis of the fee included and excluded from 6% fee limitation as compared to the estimated Construction Budget.
3. Reproduction and delivery costs of Design Review Material and other direct costs, such as travel costs, are included in the costs for each design phase and are not reimbursable. The costs of Bidding reproduction, delivery and other direct costs in accordance with the “Milestone Submittal Requirements” are included in the CD Phase. Unused site visits will be deducted at the end of the contract.
4. Provide an outline of the Quality Assurance/Quality Control actions that your firm will take during the design process and construction period. Monthly QA/QC reports are required, including actions taken to maintain project quality, schedule and budget, and will be submitted with project invoices using the NCO 15 invoice format.
5. Correction of A/E’s Errors and Omissions, including required site visits to ascertain remedial action, is included in scope of services at no additional cost to the VA.

V. REPORTS, DRAWINGS, SPECIFICATIONS AND SCHEDULES:

1. Kick-Off Meeting:
 - a. An initial kickoff meeting will be held with the Contracting Officer and local staff. This meeting will be intended to provide an onsite familiarization with the project. The VA will provide workload data and as-built drawings.
 - b. A/E Design Submissions shall comply with the VA A/E Design Submission Requirements as found in the VA Technical Information Library (TIL). The VA will provide comments and guidance following each design review session. All meetings shall be held at Marion VAMC Campus, 2401 West Main Street, Marion, Illinois 62959.
2. Schematic Design:
 - a. The A/E firm shall conduct an onsite review of the existing locations of the various functions that will be relocated to and/or remodeled within the space to help understand the needs of the departments. They shall meet with the representatives of these departments to review and compare the VA design guides to the specific departmental needs. This will serve as an opportunity for the A/E to gain a further understanding of the direction the facility wants the project to go.
 - b. After meeting with the departments, the A/E shall prepare a minimum of three separate schematic layouts of the space for review. The A/E shall provide pros and cons for each option. The A/E shall present these options to the COR who will in turn present them to the departmental leads and senior leadership. The VA will select the preferred option for the design. Space layouts shall meet current VA Space Planning Criteria’s (7610 Building Design Guide) and be in compliance with the Uniform Federal Accessibility Standards (UFAS), and VA Barrier Free Design Guide (PG-18-13).

- c. At the Schematic Design Review Meeting, the A/E should be prepared to discuss the scope of the project, provide examples of current VA standard room layouts, site drawings identifying the project area showing a basic schematic layout of the various functions, a timeline for design and a brief construction summary. The A/E shall record the minutes of the meeting to document the thoughts and concerns of the facility.

3. Design Development:

- a. The A/E firm shall make investigations necessary to thoroughly evaluate the conditions of the existing space. There will not be a systems redesign for this process. The design shall include any energy efficiency improvements available for the application. All designs shall be in accordance with A/E Submission Instructions for Minor and NRM Construction Program, PG-18-15, Volume C.
- b. Mechanical, plumbing, electrical, and fire and safety systems must meet current codes and standards, including VA Standards. All new space and utilities will be designed to meet the Interagency Security Committee's and other relevant physical security guidelines. The completed project shall be capable of achieving LEEDS certification.

4. Construction Documents:

- a. The A/E firm shall complete such further investigations to develop the accepted design development documents into working drawings and specifications for issuance through the competitive bid process.
- b. The A/E firm shall provide a complete construction plan detailing all phases and sequencing of construction to minimize the impact on the facility.
- c. In most instances, the VA Master Construction Specifications will be edited and incorporated in the project specifications. However, in those instances where corresponding VA Master Construction Specifications are not available, the A/E shall develop the specification for incorporation into the project specifications. It is required that the A/E design in accordance to VA guidelines as these guidelines supersede any other industry guidelines. It is important to note that all specification sections are to be thoroughly edited to maintain uniformity of formatting and structure. All spec writer notes must be remove.
- d. Working drawings and specifications shall provide for construction within the National Fire Protection Association Code, VA Standards, and other applicable national codes.
- e. The A/E firm shall provide a clause in the specifications indicating method by which a contractor will provide information to the firm to allow the firm to provide as-built drawings upon completion of construction.

5. As-Built Drawings:

- a. The A/E firm shall submit one (1) complete set of drawings provided on disc using AutoCAD Release 2017 or later version and also .pdf format. Models should be provided in Revit 2017 or later. The A/E shall follow the VA Standard CAD format for layering in accordance with the "B3 DESIGN CRITERIA" section of this Supplement B.

- b. The A/E firm shall submit said As-Built drawings within **thirty (30)** days of project acceptance.
- 6. Reproduction of Drawings and Specifications:
 - a. Shall be prepared in MS Word and drawings shall be prepared in MS Word and drawings shall be prepared in AutoCad 2017 (or newer) and Revit and submitted to the VA in Revit formatted files in accordance with the “B4 DESIGN DELIVERABLES” section of this Supplement B.
 - b. The A/E is required to prepare architectural drawings in Revit. One DVD/CD-ROM computer disk containing all reports, specifications, documents and drawings will be required for each milestone submission. The final Construction Document submission shall include copies of the original digital files (in *.dwg and *.rvt) in addition to the .pdf format files. All necessary reports, cost estimates and other documents shall be prepared in accordance with all applicable portions of the A/E Submission Instructions, Program Guide PG-08-15, and Volume C. The VA reserves the right to require immediate delivery of the A/E’s digital files to the VA at any time.
 - c. Specifications should be provided in both hardcopy and electronic form.
 - d. One half size set of drawings and one set of specs shall be sent to the Contracting Officer at the 90% and 100% construction document review periods.
 - e. At the 90% and 100% construction document review periods, the specifications should have all revisions incorporated providing a “clean” set of specifications for review and approval.

VI. PROMOTION OF FULL AND OPEN COMPETITION:

- 1. A/E agrees to prepare specifications that are not competitively restrictive or proprietary. Specifications will be based on performance criteria when possible and will not use a product by a specific manufacturer, except as a basis of design.
- 2. If an occasion arises that use of a product by a specific manufacturer is necessary, the A/E will provide a justification including:
 - a. Why the particular manufacturer/product must be used or why a performance specification cannot be written (proprietary design, etc.).
 - b. The evaluation criteria of alternate manufacturer’s product that must be met to be considered equal. These evaluation criteria will be used to accept the alternate manufacturer’s product as equal to the specified product. These evaluation criteria shall be comprehensive to ensure an equal product is supplied meeting contract requirements.

- c. In the event it is necessary to pre-qualify major components, the A/E will advise the Contracting Officer of such, so that the necessary action can be taken to establish a Qualified Products List.
- d. In addition to requirements of these special provisions, a certification will be provided stating the specification solicitation is not restrictive; it has been written to provide full and open competition; and will list at least three manufacturers who can supply the specified major components or systems of the project.

VII. MISCELLANEOUS:

1. The existing drawings for campus within the limits of this project will be provided to the A/E consultant For Reference Only. As-Built and Record Drawings have been found to not show the actual location, quantity, etc., of all building features and should only be used as a general guide. The A/E shall verify the existing drawings to the extent necessary for completion of construction. Files in *.dwg or *.rvt format will be provided to the A/E if those files are available for distribution.
2. The VA COR will assist the A/E when site inspection procedures impact the functioning of any of the building utility systems, services or health care operations. Spaces with controlled access will require a member of the VA staff to continuously escort the A/E Consultant whilst accessing the space. As a result, the A/E must coordinate with Marion VAMC Engineering Service to identify where the A/E is intending to perform site visit surveys at least **fourteen (14)** days in advance of any site visit.
3. Some utility systems, services or health care operations are critical, and if a disruption is required, it must be executed outside of normal working hours. Normal working hours are Monday through Friday, 7:00 a.m. to 4:30 p.m. This schedule must be coordinated with the VA COR **fourteen (14)** days in advance.
4. The A/E is responsible for recording the minutes of all meetings with the VA. The format of the meeting minutes shall be subject to prior approval from the VA COR. The draft meeting minutes shall be distributed for review and comment within one working day following the meeting to the COR, who will in turn distribute the same to each invited VA-attendee as required. Meeting minutes are loaded to Submittal Exchange or equal web system. The VA shall have two working days upon receipt of the meeting draft documents to make any additional comments or corrections. The final meeting minutes shall be issued within three working days after the meeting to the VA CO and the COR.
5. Coordination of each submittal will be fully documented by the A/E. Using industry standard methods, the A/E will collect, document and resolve all comments and consolidate them into a usable format. The resolution of each comment requires the COR and commenter's concurrence. For each design submittal, the A/E will conduct a comment review meeting where each comment received will be either:
 - a. Concurred as Written
 - b. Non-Concurred; Request Discussion

c. Requires Clarification

6. At the conclusion of the comment review meeting, the A/E will provide meeting minutes showing the status of each comment made. The resolution will be incorporated into the project documents (drawings, specifications, etc.) prior to the next submittal. Any comment resolutions not included in the project documents will be provided to the CO and COR with explanation as to why each was not included.

VIII. DESIGN FUNDING LIMITATION:

1. In accordance with FAR 52.236-22, Design within Funding Limitation (APR 1984), the estimated Total Construction Contract Price for this Project proceeding into CDs is not to exceed Construction budget listed above not including any Project Contingencies, VA Furnished/Installed Medical Equipment, VA Furnished/Installed FFE or Professional Service Fees. The A/E is responsible for designing the Project within the estimated total Construction Contract Price for the Design Development and Construction Document phases of this Project. If, at any time, the A/E determines that the design's construction cost (excluding contingencies) will exceed the Total Construction Contract Price (TCCP) the A/E shall immediately inform the Contracting Officer in writing. Such written notification shall include the reasons why the A/E believes the TCCP will probably be exceeded and shall offer proposed solutions to reduce the project cost to be within the TCCP.
2. Deductive alternates (acceptable to the VA) of up to 20% of the Total Construction Contract Price shall be included in the Contract Documents. These Deductive alternatives shall not be utilized to lower the estimated Total Construction Contract Price. The base design must be fully functional with or without selection of the alternates.

IX. PROBABLE CONSTRUCTION COST ESTIMATE:

1. The estimates shall be prepared using VA Form 10-6238 or equivalent, Estimate Worksheet and provide sufficient detail to identify the cost of project systems and components as appropriate for the stage of development of the design in the SD, DD and CD Phases. Update construction cost data with each submittal review milestone. Costs presented shall be broken down sufficiently to allow for direct comparison cost with the contractor's proposed cost.
2. VA Form 10-6238 or equivalent, Estimate Worksheet: This form serves two purposes:
 - a. To show the division of cost between labor and material for each cost item; and
 - b. To extend the level of detail of an estimate beyond square foot cost summaries.

The estimated costs shall include all labor, materials, mobilization, overhead, insurance, and applicable contractor markup. A/E may utilize their estimate worksheets if approved in advance by the VA.

3. Electronic copy of estimate shall be provided with each submittal. Estimate detail shall include a breakdown by Project specification division number for labor and material broken out by quantity and unit cost for each type of material or equipment, including equipment rentals.

General Division shall include costs associated with project duration. Include Summary of estimate by Division Number with each submittal.

4. Providing the estimates as "lump sum" or "allowance" figures in the Construction Documents will not be acceptable. For the Schematic estimate, a 15% total contingency allowance is allowable. For the Design Development estimate, a 10% total contingency allowance is allowable. Contingency allowances are not permitted in the final Construction Documents Estimate. Construction cost estimating shall correspond with the work shown in the drawings and be commensurate with the degree of detail shown in the drawings. Construction cost estimating shall also include all elements within the contractor's bid; such as reasonable overhead and profit and fees across all tiers, insurance, taxes, bonds, hazardous abatement and any other such items. Costs will include itemization of the following:
 - a. Number of labor hours by trade (or crew¹)
 - b. Labor dollars by trade (or crew¹)
 - c. Material quantity
 - d. Material dollars
 - e. Equipment Rental and associated cost
 - f. Any other Direct cost not associated with work to be performed by a specific trade

¹ When using a crew, instead of a trade, then use a crew code and include a schedule, or legend, of only the crew codes applied, showing each type of trade, each hourly rate, and each trade person quantity.

5. Some costs for General Conditions or General Requirements may also be required in the project estimate whenever these costs can be shown to be direct costs as opposed to indirect (overhead) costs. For example, if it is common practice for the Contractor to use a rented negative air HEPA machine and consumable plastic sheeting to comply with the Infection Control Risk Assessment (ICRA) requirements, then the associated costs would be direct material and labor costs. In addition, if the Construction Documents require the Contractor to perform fulltime on-site superintendence solely dedicated to the construction project, then these would be direct labor costs. Furthermore, whenever the Contractor is to provide on-site factory-authorized training to the VA staff, then the travel, lodging, meals and incidental costs would be direct costs. If the Construction Documents require the Contractor to perform housekeeping services, then the associated costs would be direct costs. If the Construction Documents call the Contractor to execute specific safety and OSHA activities, then the costs associated with these activities would be direct labor and material costs; however, OSHA activities performed by the Contractor by its staff at its home office and not specified in the Construction Documents, then such associated costs would be indirect costs and would not be included with the project estimate.

B2 INDUSTRIAL HYGIENE SERVICES:

I. INDUSTRIAL HYGIENIST TASKS:

For the work scope described in B1 SCHEDULE OF SERVICES above Hazardous materials (Asbestos, Lead, Mold, heavy metal and etc...) **ARE** included in this task order amendment. Lead and Asbestos Hazardous Material Inventories within the project limits may be available from the VA for consultant's reference.

1. Architect/Engineer Responsibilities:

- a. Determine if asbestos, lead or mold is present. If found, design a method for abating/remediating all asbestos containing materials, lead containing materials and mold contaminated items impacting this project.
- b. Determine location of asbestos, lead and/or mold contamination, and any water and moisture damage. Work with the I.H. to uncover and assess hidden areas.
- c. For mold contamination, determine entry point of water infiltration
- d. For asbestos and lead contamination, determine the extent of abatement required to eliminate all impacts to this project.
- e. Review the I.H. Report.
- f. Produce a written report compiling all necessary information.

II. VA PROFESSIONAL INDUSTRIAL HYGIENIST (VPIH):

As required, the A/E will retain a specialist in VPIH services to provide sufficient oversight of the abatement process. The following VPIH services will comply with federal, state and local laws and regulations:

1. Review of Contractor Provided Asbestos Abatement Plan: The VPIH is responsible for reviewing and approving the Contractor's Professional Industrial Hygienist (CPIH) work plans. Provide comments and recommendations to ensure VA personnel and/or patients are not exposed to asbestos fibers.
2. Provide Air Sampling: Provide air sampling of the construction site to verify that no asbestos fibers were released during the asbestos abatement activities. Field work and sampling activities shall be performed by trained personnel under the direction of a licensed Industrial Hygienist or Board Certified Microbial Consultant.
3. Provide Abatement Oversight: Provide oversight of the abatement contractor to ensure compliance with abatement laws and regulations.
4. Provide Final Clearance: Ensure the contractor has completed the abatement work required and provide final clearance that the abatement work is complete and there are no impacts to the Medical Center.
5. Provide Final Report: Provide a final report showing daily logs, sampling results and collected documents.

III. All work shall be in compliance with applicable laws, regulations and procedures published or issued by federal, state and local regulatory agencies for the abatement and control of hazardous materials.

B3 GOVERNMENT FURNISHED DESIGN CRITERIA:

- I.** The A/E shall design in accordance with the requirements of the following VA publications, which can be found in the VA [Tech Info Library](http://www.cfm.va.gov/TIL/) (TIL) at <http://www.cfm.va.gov/TIL/>:

- VA Master Construction Specifications, Program Guide, PG-18-1;
<http://www.cfm.va.gov/TIL/spec.asp>
- VA Design and Construction Procedures, PG-18-3; <http://www.cfm.va.gov/TIL/cPro.asp>
- VA Standard Details, PG-18-4, Volumes 1, 3, 4, 5, 6 & 7; <http://www.cfm.va.gov/TIL/sDetail.asp>
- VA Equipment Guide List, PG-18-5; <http://www.cfm.va.gov/TIL/equip.asp>
- VA Physical Security and Resiliency Design Manual,
<https://www.cfm.va.gov/til/PhysicalSecurity/dmPhySec.pdf>
- VA Seismic Design Handbook, H-18-8; <http://www.cfm.va.gov/TIL/seismic.asp>
- VA Space Planning Criteria, PG-18-9; <http://www.cfm.va.gov/TIL/planning.asp>
- VA Design Manuals for the applicable disciplines, PG-18-10; <http://www.cfm.va.gov/til/dManual.asp>
- VA Design Guides for graphics, PG-18-12; <http://www.cfm.va.gov/til/dGuide.asp>
- VA Barrier Free Design Guide, PG-18-13; <http://www.cfm.va.gov/TIL/accessibility.asp>
- VA Room Finishes, Door and Hardware Schedule, PG-18-14;
<http://www.cfm.va.gov/TIL/room/RoomFinishes.pdf>
- VA Minimum Requirements for A/E Submissions, PG-18-15;
<http://www.cfm.va.gov/til/aeDesSubReq.asp>
- VA Environmental Compliance Manual, PG-18-17; <http://www.cfm.va.gov/til/spclRqmts.asp#EC>
- VA Design Alerts and Quality Alerts; <http://www.cfm.va.gov/til/alert.asp>
- VA CAD Standards; <http://www.cfm.va.gov/til/projReq.asp#cad>
- VA Design Review Checklists; <http://www.cfm.va.gov/til/projReq.asp#reviewCheck>
- VA AE Design Submission Requirements; <http://www.cfm.va.gov/til/aeDesSubReq.asp>
- Office of Construction & Facilities Management Design Manual, Architectural Design Manual,
<http://www.cfm.va.gov/til/dManual/dmARhosp.pdf>
- Seismic Design Manual for VA Retrofit Projects
- VA Directive 0056 – VA Sustainable Building Program
- VA Medical Center Memorandums (“MCM”) (Request From COR)

II. Other design criteria may be required to meet specific requirements of the design work scope that are not listed above, or are not available on the VA Office of Facilities web site. The A/E is responsible for obtaining and complying with all such criteria.

B4 DESIGN DELIVERABLES:

I. DISTRIBUTION OF A/E MATERIALS TO THE VA:

The A/E shall provide the means necessary to submit large electronic files to VA recipients. Files over 10 MB may not be transmitted through the VA email system. File transfer protocol websites (FTP) are prohibited for use by VA employees. Provide Submittal Exchange or equal web system. Service should be provided through construction period services and allow use by construction contractor.

1. Reproduction and delivery costs of Design Review Material in accordance with the “Milestone Submittal Requirements” are included in the costs for each design phase and are not reimbursable.

2. Bidding and Construction Material reproduction and delivery costs in accordance with the “Milestone Submittal Requirements” are included in the CD Phase and are not reimbursable.
3. The A/E shall provide a “Stand Alone Certification Memo” to the VA with the **60%** Contract Documents for Bidding. (see attached format).
4. Record Documents (“As-Built”) shall be submitted to the VA immediately prior to final acceptance of the project by the VA. Record Documents shall conform to the VA standard requirements including originals and record specifications incorporating all of the actual finished conditions, project revisions/amendments and products. Record Document Material reproduction and delivery costs in accordance with the “Milestone Submittal Requirements” are included in the CPS Phase and are not reimbursable.

II. MILESTONE SUBMITTAL REQUIREMENTS (Refer to following page):

MILESTONE SUBMITTAL REQUIREMENTS						
Number of Required Submittal Documents						
Submittal Description	Digital File		Letter	Drawings		
	.orig	.pdf		Half Size	Full Size	Mylar
Design Narrative Submittal:						
Condition Assessment		1				
Cost Estimate	1	1				
30% Schematic Design Submittal:						
Cost Estimate	1	1				
Reports & Calculations		1				
Outline Specifications	1	1				
Drawings		1		2	1	
60% Contract Document Submittal						
Cost Estimate	1	1				
Reports & Calculations		1				
Draft Specifications	1	1				
Drawings		1		2	1	
Stand Alone Certification Memo	1	1				
90% Contract Document Submittal						
Cost Estimate	1	1				
Reports & Calculations		1				
Specifications	1	1				
Drawings		1		2	1	
100% Contract Documents for Bidding						
Final Cost Estimate	1	1				
Reports & Calculations		1				
Specifications	1	1				
Drawings		1		2	1	
Release for Construction Set (Incorporate all Solicitation Amendments):						
Final Cost Estimate	1	1				
Reports & Calculations		1				
Specifications	1	1				
Drawings		1		2	1	
Existing Condition Record Drawings		1		2	1	
Record Document Submittal:						
Final Project Construction Cost	1	1				
Operation & Maintenance Manuals	1	1				
Record Specifications	1	1				
Record Drawings		1		1	1	

Note 1: ".orig" indicates digital file in the document's original file format i.e. ".doc", ".rvt", ".dwg", ".xls", etc.

Note 2: Letter size reports may contain 11" x 17" drawing or exhibit pages tri-folded to letter size as required.

Note 3: Letter size (8.5" x 11") submittal documents less than or equal to 25 pages may be stapled, all other letter size documents shall be bound with a Cover Sheet containing the Project information, date and Table of Contents.

B5 DESIGN REVIEW AND COMPLETION SCHEDULE:

- I.** The A/E shall perform the work required by this contract within **365 calendar days** reflective of schedule below. The A/E shall provide Professional Architects and Engineers as directed by the Contracting Officer (CO) who are familiar with the work to attend the Design Reviews on the following days:

Design Phase	Phase Duration (Days¹)	Calendar Days from NTP
Beginning of Site Investigation	20	20
Completion of Site Investigation	54	74
Deliver Conceptual Study and Scope to CO	22	96
Deliver Schematic Design (30%) Materials to CO	33	129
Review Schematic Design (30%) Materials	22	151
Deliver Design Development (60%) Materials to CO	44	195
Review Design Development (60%) Materials	22	217
Deliver Design Development (90%) Materials to CO	43	260
Review Design Development (90%) Materials	22	282
Deliver Construction Documents (100%) to CO	26	308
Review Construction Documents (100%)	29	337
Deliver Final Contract Documents to CO for Bidding	28	365
Record Document Submittal:	TBD	TBD

Note 1: Number of Days is calendar days.

- The delivery of the conceptual study and scope shall coincide with a conceptual review meeting at the Marion VAMC Engineering Building Conference Room. The AE shall collect all design options, data and illustrations and present them to the CO and COR for evaluation. The engineering building white board and projector will be available for use by the AE but the AE must supply their own computer. The CO, COR and all other meeting attendees will review the proposed design options presented and select an option to move forward with. The decision on the design option selection will be made by the VA and returned to the architect within 24 hours of the conceptual review meeting.

2. The Contracting Officer (CO) may schedule additional Design Reviews as necessary when it is determined by the CO that such reviews are required for satisfactory completion of A/E contract requirements. The A/E will be reimbursed for additional review meetings in accordance with the rate sheet, unless it is solely determined by the CO that such additional reviews were requested by the A/E, or were required due to the A/E's failure to meet contract requirements.
3. The A/E is responsible for recording the minutes of all meetings. The format of the meeting minutes shall be subject to prior approval from the VA COR. The draft meeting minutes shall be distributed for review and comment within one working day following the meeting to the COR, who will in turn distribute the same to each invited VA-attendee as required. Notification by email to the COR that the draft documents are available on an A/E provided ftp site, or as an attachment, is the preferred method. The VA shall have two working days upon receipt of the meeting draft documents to make any additional comments or corrections. The final meeting minutes shall be issued within three working days after the meeting to the VA CO and the COR.
4. Milestone review minutes shall comply with the requirements for meeting minutes, except that two days shall be added to each response period. In addition, the submitted milestone review drawing mark-ups by the VA and any review meeting comments shall be described in a tabular form with the comment (listed by discipline), comment originator, partly responsible for responding to comment, and a column indicating whether comment was incorporated into design documents. If comment was not incorporated or only partially incorporated, provide an explanation for the action.
5. The interim and final documents (SD, DD and CD) from this Task Order will be subject to review and approval (by signature) of all members of an Integrated Project Team ("IPT") consisting of Medical Center personnel. These individuals will be identified by the COR by the first design review meeting. The A/E shall prepare a title sheet for each submission and phase as appropriate to receive approval signatures from members of the IPT. The COR will provide A/E with a copy of the Memorandum identifying IPT members.

B6 CONCEPTUAL STUDY AND SCOPE SUBMISSION:

I. ARCHITECTURAL, STRUCTURAL, MECHANICAL AND ELECTRICAL:

1. Submit 2-D and 3-D sketches as necessary to detail the proposed design conceptual upgrades/modifications and the associated extents of work. Include other pertinent information as necessary to allow Medical Center Leadership, COR's and End-Users to choose which concept they desire to be developed further.
2. Identify existing system code compliance issues and the options available to address those issues.
3. Show schematic sections, as needed, to define the building and/or equipment concepts. Include field verified locations of existing or remaining details.
4. Show roof plans if required.

5. Show legends that as a minimum reflect all items denoted on drawing at this submission.

II. COST ESTIMATES:

1. A/E shall submit construction cost estimates for each conceptual configuration simultaneously with the drawings. The estimates shall reflect, as accurately as possible, the total construction cost associated with the different concepts.

B7 SCHEMATIC REVIEW SUBMISSION (30% REVIEW):

I. SPECIFICATIONS:

1. Submit an index of all VA Master Construction Specification sections intended for use. Include a brief description of any A/E originated sections to be used.
2. A/E shall obtain the VA Master Construction Specifications and any other VA Technical Documents at <http://www.va.gov/facmgt>.

II. ARCHITECTURAL:

1. Submit as a minimum, a layout for each project. Include fixed equipment at a scale not less than 1/8" = 1' 0". Submit 1/4" = 1' 0" scale floor plans of special rooms such as the Cath Lab procedure room, recovery room, etc., to show layout and identification of equipment. Include elevation views and or renderings as necessary to allow end users to decide which concept they desire to be developed.
2. Show schematic sections, as needed, to define the building and/or equipment concepts. Include field verified locations of existing or remaining details.
3. Show roof and penthouse plans if required.
4. Show legends that as a minimum reflect all items denoted on drawing at this submission.

III. STRUCTURAL:

1. Show the selected structural system for new procedures, if required. Show beam depths, slab thickness, column sizes and expansion joints.
2. Submit supporting calculations.

IV. MECHANICAL:

1. Submit, as a minimum, outline plan drawings showing HVAC replacement unit locations, sizes of connection lines to existing services, and any lines to be removed.

2. Provide a written description of the replacement HVAC unit considered for the project and how it will be connected to the existing system to justify final system selection.

V. PLUMBING:

1. Submit, as a minimum, outline plan drawings showing main water line locations feeding the project boundary, compressor locations, sizes of main lines , including drains, points of connection to existing piping, all new piping and piping to be removed.

VI. ASBESTOS, LEAD AND MOLD:

1. If any areas are identified as containing hazardous materials, show areas of suspected Asbestos Containing Material (ACM), lead and mold abatement required. Show quantities that will require abatement for each. Lead may be lead paint, lead shower pans, lead lined walls, lead lined doors, etc.

VII. ELECTRICAL:

1. Submit, as a minimum, floor plan showing new and existing panel locations and sizes. Indicate point of connection to existing distribution centers.

VIII. PHASING:

1. Submit, as a minimum, drawing number indicating a separate phasing drawing.

IX. FIRE PROTECTION AND LIFE SAFETY:

1. Show fire protection and life safety plans.

X. COST ESTIMATES:

1. A/E shall submit a construction cost estimate simultaneously with the drawings for this project at each project submission.
2. This estimate shall show the cost of construction which would be expected to be reflected by the Construction Contractor's bids, if the bids were submitted on the same date as the estimate. The level of detail for this estimate shall be consistent with the degree of completeness of the drawings being submitted. Simply stated, this means that if a construction element is shown, it must be priced; if it is shown in detail, it must be priced in detail. For detailed elements, "lump sum" or "allowance" figures will not be acceptable.
3. The detailed estimate will contain unit cost, quantity, and total cost for both materials and labor. Each division (architectural, electrical, mechanical, etc.) will be broken down in the above manner.
4. All costs shown shall include all labor insurance and taxes, sales taxes on material, and applicable upper and lower tier Contractor and Sub-Contractor markup.

5. For the first review only include a contingency allowance of 15%.
6. When union labor is used in the cost estimate, the A/E shall include welfare and pension fund costs in labor units as appropriate.
7. Show a sub-total of Labor and Material for each sub-heading.
8. Show sales tax as a percentage of material.

B8 DESIGN DEVELOPMENT SUBMITTAL (60% REVIEW):

I. SPECIFICATIONS:

1. Submit bound and edited VA Master Specifications of each discipline for review. Include a brief description of any A/E originated sections to be used. A copy of the 30% specification showing changes that have been made shall be submitted.
2. Submit an updated list of all the submittals that the contractor will be required to provide. Provide a draft submittal register.

II. ARCHITECTURAL:

1. Submit near complete floor plans and half complete details, schedules and large-scale plans. The architectural working drawings should be 60% complete.
2. Show room titles and numbers, door sizes, types and swing on the 1/8" scale plans. Dimension corridors and stairs. Indicate and identify any portable equipment and plumbing fixtures.
3. Draw exterior elevations at 1/8" scale unless building is of such size that a complete facade cannot be shown on the standard size sheets, in which case draw the elevations at 1/16" scale. Show story heights, penetrations, materials, penthouses, architectural screens, roof enclosures, skylights, and stacks on elevations as applicable.
4. Show room finish symbols on the 1/8" scale floor plans only and key them to the room finish schedule in section 09 06 00 of the specifications.
5. Show general notes, door, and window schedules. Include other scheduled information.
6. Indicate all fire partitions, smoke partitions, and safety protective elements (including lead linings and radio frequency shielding).
7. Draw reflected ceiling plans of typical areas and special rooms, indicated all ceiling mounted equipment, lighting fixtures, air diffusers, registers, etc.

8. Where existing spaces occur that require refinishing, a survey shall be prepared to include color, materials, furnishings, and window treatments. Project Engineer will work with A/E to coordinate interior design and color.
9. Submit a 60% completed color design containing selected colors, textures, and materials listed in the proper format. This shall include manufacturers by name, number, and pattern. They will be coded and located for both interior and exterior.

III. STRUCTURAL:

1. Submit drawings to scale showing tentative sizes of columns, beams, braces and slabs if structural reinforcing is required. The structural drawings shall be complete enough to correlate with architectural and mechanical features. Submit calculations developed to date.
2. Show sections and details so that construction features can be defined.

IV. MECHANICAL:

1. Provide complete load and sizing calculations for all HVAC, including sound attenuation, energy conservation, and steam distribution systems and equipment. Provide psychrometric analysis of each air-handling unit. If computerized methods are used, provide supplemental sketches and explanation of terms and abbreviations.
2. Provide 1/8" scale floor plans showing the HVAC and steam distribution systems and the equipment. Coordinate the design with other disciplines.
3. Provide flow and control diagrams. Include on the diagrams, where applicable, air conditioning, refrigeration systems, steam, condensate, conditioned air systems, exhaust air systems, hot water, chilled water, and condenser water. Provide sequence of operations and controls for all HVAC systems and equipment. Diagrams shall be complete with legend and description of operations.
4. Provide complete design for fire and safety systems that are associated with the HVAC systems. Show the fire and safety design on the HVAC plans and control diagrams. Show smoke and fire partitions and related dampers.
5. Provide complete equipment schedule drawings reflecting the sizing calculations of all major equipment, such as chillers, refrigeration elements, cooling towers, pumps, fans and air handling units. Submit calculations for fan pressures and pump heads.
6. Provide 1/4" scale plans for all mechanical equipment rooms. Provide 1/4" scale sections through each floor in typical locations for any tight spots and mechanical equipment rooms. Show in the 1/4" scale drawings of all equipment, piping, ductwork, and other interfacing elements. Provide adequate space for installation, access and maintenance of equipment.

7. In the calculations provide a tabular recapitulation indicating the air balance between the supply and exhaust in the design computations showing the volume of air, room names and numbers. Show that individual areas that normally constitute an integrally balanced system, such as surgery, administration, etc., are in balance and do not adversely affect the balance of the entire building. For air conditioning systems, added to an existing building, show the air balance with respect to the existing exhaust and supply systems to provide a complete building study.
8. For all existing buildings where modification of existing air, water (hot and chilled) or steam is required, show riser diagrams with points of interface of existing and new work.
9. Provide demolition drawings, showing the extent of removal for existing systems and equipment.
10. Show room names on HVAC plans so that compliance with criteria can be checked.
11. Submit any changes that will affect economic studies.
12. On drawings, show double line representation of all ducts, single line representation of piping systems 6" or smaller, and double line representation of piping systems 8" or larger for heating, air conditioning, and ventilation the building. Clearly indicate the sizes of ducts and piping. Locate fire and smoke dampers.
13. Determine probable steam demand for heating, ventilating and humidification.
14. Major details, elevations, sections, and legends on the drawings are to be completed. Show a section of each critical point in corridors, indicating the location and space allocation intended for all ducts, piping, and larger conduits regardless of trade. These drawings shall be three-dimensional to show how the ductwork, piping and conduit are to be installed in the space above ceiling. Show typical sections for entire floor, where applicable.
15. This submittal shall have all the HVAC systems substantially completed.

V. PLUMBING:

1. Show and identify all plumbing fixtures and oxygen, medical compressed air, and vacuum outlets on 1/8" = 1' 0" scale floor plans. Plumbing and sanitation plans shall be complete, including sizing. Submit calculations.

VI. ASBESTOS, LEAD AND MOLD:

1. Show areas that require asbestos and lead abatement and the approximate quantities that will require abatement for each, i.e. pipe insulation, ceiling tile, floor tile, lead paint, lead lined walls, lead lined doors, etc.
2. Present phasing plans if abatement is in multiple areas. Also, indicate if off-hour work will be required in each phase.

3. Indicate on drawings and specifications, approximate materials required for replacement of abated ACM, i.e. pipe insulation, etc.

VII. ELECTRICAL:

1. Provide sheet for symbols.
2. Submit finalized one-line and riser diagrams of electrical power distribution systems and the emergency power systems. (Final equipment rating may vary, but all equipment should be located, sized, and identified). Submit final load calculations and panel board sizes.
3. All floor plans shall have room titles and area function shown on this review set of drawings. Location of all lighting fixtures, outlets for power, fire alarm devices, and signal outlets, and devices shall be shown. Layouts of specialty areas (radiology, office/exam, bed areas, O.R.'s ICU's etc.) should be substantially laid out at this review.
4. All electrical drawings shall show smoke partitions and fire alarm zones.
5. Submit lighting calculations, typical short-circuit and voltage drop calculations.
6. Provide reflected ceiling plans showing lighting layout, location of all duct openings, and area smoke detectors.
7. Show locations and 60% complete details of essential electrical system changes and upgrading.
8. Indicate sized feeders and primary transformers with options.
9. Developed single line diagram against field gathered data.

VIII. PHASING:

1. Present proposed phasing plans for alteration projects if construction area will be vacated piecemeal or certain rooms within the construction contract need to be in operation. Phasing information shall be complete and in written or chart form and will have a separate phasing sheet in the drawing set.

IX. FIRE PROTECTION AND LIFE SAFETY:

1. Show fire protection and life safety plans.

X. COST ESTIMATES:

1. A/E shall submit a construction cost estimate simultaneously with the drawings for this project at each project submission.

2. This estimate shall show the cost of construction which would be expected to be reflected by the Construction Contractor's bids, if the bids were submitted on the same date as the estimate. The level of detail for this estimate shall be consistent with the degree of completeness of the drawings being submitted. Simply stated, this means that if a construction element is shown, it must be priced; if it is shown in detail, it must be priced in detail. For detailed elements, "lump sum" or "allowance" figures will not be acceptable.

B9 DESIGN DEVELOPMENT SUBMITTAL (90% REVIEW):

I. SPECIFICATIONS:

1. Submit bound and edited VA Master Specifications of each discipline for review. Include a brief description of any A/E originated sections to be used. A copy of the 60% specification showing changes that have been made shall be submitted.
2. Submit an updated list of all the submittals that the contractor will be required to provide. Provide a draft submittal register.

II. ARCHITECTURAL:

1. Submit near complete floor plans and half complete details, schedules, and large-scale plans. The architectural working drawings should be 90% complete.
2. Show room titles and numbers, door sizes, types and swing on the 1/8" scale plans. Dimension corridors and stairs. Indicate and identify any portable equipment and plumbing fixtures.
3. Draw exterior elevations at 1/8" scale unless building is of such size that a complete façade cannot be shown on the standard size sheets, in which case draw the elevations at 1/16" scale. Show story heights, penetrations, materials, penthouses, architectural screens, roof enclosures, skylights, and stacks on elevations as applicable.
4. Show room finish symbols on the 1/8" scale floor plans only and key them to the room finish schedule in section 09 06 00 of the specifications.
5. Show general notes, door, and window schedules. Include other scheduled information.
6. Indicate all fire partitions, smoke partitions, and safety protective elements (including lead linings and radio frequency shielding).
7. Draw reflected ceiling plans of typical areas and special rooms, indicated all ceiling mounted equipment, lighting fixtures, air diffusers, registers, etc.
8. Where existing spaces occur that require refinishing, a survey shall be prepared to include color, materials, furnishings, and window treatments. VA interior designer will work with A/E to coordinate interior design and color.

9. Submit a 90% completed color design containing selected colors, textures, and materials listed in the proper format. This shall include manufacturers by name, number, and pattern. They will be coded and located for both interior and exterior.
10. Precisely and clearly illustrate the development and refinements of the interior design. This may be sketches, illustrations, and/or colored floor plans for typical interior areas. Submit one paste-up board, which may be returned as required.

III. STRUCTURAL:

1. Submit drawings to scale showing tentative sizes of columns, beams, braces and slabs if structural reinforcing is required. The structural drawings shall be complete enough to correlate with architectural and mechanical features. Submit calculations developed to date.
2. Show sections and details so that construction features can be defined.

IV. MECHANICAL:

1. Provide complete load and sizing calculations for all HVAC, including sound attenuation, energy conservation, and steam distribution systems and equipment. Provide psychrometric analysis of each air-handling unit. If computerized methods are used, provide supplemental sketches and explanation of terms and abbreviations.
2. Provide 1/8" scale floor plans showing the HVAC and steam distribution systems and the equipment. Coordinate the design with other disciplines.
3. Provide flow and control diagrams. Include on the diagrams, where applicable, air conditioning, refrigeration systems, steam, condensate, conditioned air systems, exhaust air systems, hot water, chilled water, and condenser water. Provide sequence of operations and controls for all HVAC systems and equipment. Diagrams shall be complete with legend and description of operations.
4. Provide complete design for fire and safety systems that are associated with the HVAC systems. Show the fire and safety design on the HVAC plans and control diagrams. Show smoke and fire partitions and related dampers.
5. Provide complete equipment schedule drawings reflecting the sizing calculations of all major equipment, such as chillers, refrigeration elements, cooling towers, pumps, fans and air handling units. Submit calculations for fan pressures and pump heads.
6. Provide 1/4" scale plans for all mechanical equipment rooms. Provide 1/4" scale sections through each floor in typical locations for any tight spots and mechanical equipment rooms. Show in the 1/4" scale drawings of all equipment, piping, ductwork, and other interfacing elements. Provide adequate space for installation, access and maintenance of equipment.

7. In the calculations provide a tabular recapitulation indicating the air balance between the supply and exhaust in the design computations showing the volume of air, room names and numbers. Show that individual areas that normally constitute an integrally balanced system, such as surgery, administration, etc., are in balance and do not adversely affect the balance of the entire building. For air conditioning systems, added to an existing building, show the air balance with respect to the existing exhaust and supply systems to provide a complete building study.
8. For all existing buildings where modification of existing air, water (hot and chilled) or steam is required, show riser diagrams with points of interface of existing and new work.
9. Provide demolition drawings, showing the extent of removal for existing systems and equipment.
10. Provide separately identifiable drawings at 1/8" scale for asbestos abatement work.
11. Show room names on HVAC plans so that compliance with criteria can be checked.
12. Submit any changes that will affect economic studies.
13. On drawings, show double line representation of all ducts, single line representation of piping systems 6" or smaller, and double line representation of piping systems 8" or larger for heating, air conditioning, and ventilation the building. Clearly indicate the sizes of ducts and piping. Locate fire and smoke dampers.
14. Determine probable steam demand for heating, ventilating and humidification.
15. Major details, elevations, sections, and legends on the drawings are to be completed. Show a section of each critical point in corridors, indicating the location and space allocation intended for all ducts, piping, and larger conduits regardless of trade. These drawings shall be three-dimensional to show how the ductwork, piping and conduit are to be installed in the space above ceiling. Show typical sections for entire floor, where applicable.
16. This submittal shall have all the HVAC systems substantially completed.

V. PLUMBING:

1. Show and identify all plumbing fixtures and oxygen, medical compressed air, and vacuum outlets on 1/8" = 1' 0" scale floor plans. Plumbing and sanitation plans shall be complete, including sizing. Submit calculations.

VI. ASBESTOS, LEAD AND MOLD:

1. Indicate all areas that require asbestos and lead abatement and the quantities of each type of abatement, i.e. each size pipe insulation, ceiling tile, floor tile, lead paint, lead lining, lead shower pan, etc.

2. Present phasing plans if abatement is in multiple areas. Also indicate if off-hour work will be required in each phase.
3. Show possible full abatement setup if required for abatement.
4. Indicate on drawings and specifications, all materials required for replacement of abated ACM, i.e. pipe insulation, tiles.

VII. ELECTRICAL:

1. Provide sheet for symbols.
2. Submit finalized one-line and riser diagrams of electrical power distribution systems and the emergency power systems. (Final equipment rating may vary, but all equipment should be located, sized, and identified). Submit final load calculations and panel board sizes.
3. All floor plans shall have room titles and area function shown on this review set of drawings. Location of all lighting fixtures, outlets for power, fire alarm devices, and signal outlets, and devices shall be shown. Layouts of specialty areas (radiology, office/exam, bed areas, O.R.'s ICU's etc.) should be substantially laid out at this review.
4. All electrical drawings shall show smoke partitions and fire alarm zones.
5. Submit lighting calculations, typical short-circuit and voltage drop calculations.
6. Provide reflected ceiling plans showing lighting layout, location of all duct openings, and area smoke detectors.
7. Show locations and 90% complete details of essential electrical system changes and upgrading.

VIII. PHASING:

1. Present proposed phasing plans for alteration projects if construction area will be vacated piecemeal or certain rooms within the construction contract need to be in operation. Phasing information shall be complete and in written or chart form and will have a separate phasing sheet in the drawing set.

IX. FIRE PROTECTION AND LIFE SAFETY:

1. Show fire protection and life safety plans.

X. COST ESTIMATES:

1. A/E shall submit a construction cost estimate simultaneously with the drawings for this project at each project submission.

2. This estimate shall show the cost of construction which would be expected to be reflected by the Construction Contractor's bids, if the bids were submitted on the same date as the estimate. The level of detail for this estimate shall be consistent with the degree of completeness of the drawings being submitted. Simply stated, this means that if a construction element is shown, it must be priced; if it is shown in detail, it must be priced in detail. For detailed elements, "lump sum" or "allowance" figures will not be acceptable.
3. Contracting Officer requirements – in addition to submissions to the COR, the A/E shall submit **ALL** construction cost estimates directly to the Contracting Officer. The final construction cost estimate delivered with the 100% specs and drawings shall be marked "For Official Use Only" and be signed and dated by the architect. This document will be used as the Independent Government Cost Estimate (IGCE) for construction period services.
4. Use of Forms – NCO 15 Form 10-6298, for A/E Fee Proposal Submission Form.

B10 CONSTRUCTION DOCUMENT SUBMITTAL (100% REVIEW):

I. GENERAL:

1. Contract drawings and specifications and all related documents shall be completed, fully coordinated, bound and ready for reproduction for bidding and construction purposes. A copy of the 90% specification showing changes that have been made shall be submitted.
2. All submissions shall be accompanied by a transmittal letter with line item breakdown detailing specific modifications from prior submission.
3. The A/E shall deliver the 100% Final (working) Drawing set to the COR after review and approval. The final working drawings shall bear the seal of the Registered Architect, Professional Engineers, Fire Protection Engineer and Certified Industrial Hygienist responsible for the design.
4. Prior to reproduction for issue for construction bids, the A/E shall make any changes to the working drawings, calculations, and specifications identified as necessary during the Third Review.
5. Provide a Submittal Register.
6. The final design documents from the resulting task order will be subject to review and approval (by signature) of all members of an Integrated Project team (IPT) consisting of medical center personnel. These individuals will be identified by the COR at the first design review meeting.
7. Drawing cover sheet will have a signature block with IPT signatures.
8. A 1-2 page Statement of Work for the construction project will be provided in MS word to the Contracting Officer. Statement of work will reflect General Requirements and the number of calendar days for construction.

B11 CONSTRUCTION DOCUMENT SUBMITTAL (AS-BUILT):

I. GENERAL:

1. A/E firm shall provide a clause in the specifications indicating the method by which a contractor will provide information to the firm to allow the firm to provide as-built drawings upon completion of construction. As-built drawings shall not only cover work involved in the construction but shall also cover existing adjacent areas as defined within the scope of the preliminary survey.
2. A/E firm shall submit one (1) copy of prints. A complete set of drawings shall also be provided on disc using the latest version of AutoCAD used by the VA.
3. BASE Revit and AutoCAD drawings:
 - a. If the drawings already exist in Revit and AutoCAD then the contractor/architect will modify the base drawing and add their changes to the drawings such that only one drawing will result. If the existing drawings DO NOT exist already in Revit and AutoCAD then the Contractor will put their drawings in Revit and AutoCAD so as to match up with paper drawings.
 - b. Revit and AutoCAD changes should place all different disciplines on the separate layer and be to same scale. Layers should match the existing layers on the particular Revit and AutoCAD drawing used. If there is no Revit and AutoCAD drawing, then layers should match those used on the drawings.
 - c. Identify any underground objects encountered during investigation and construction for reference. Note that utilities damaged during construction shall be the responsibility of the contractor.

B12 ATTENDANCE AT CONFERENCES:

- I.** Attendance at Conceptual Study and Scope Submission conference **IS** required with regard to this task order.
- II.** Attendance at Schematic Design (30%) Review conference **IS** required with regard to this task order.
- III.** Attendance at Design Development (60%) Review conference **IS** required with regard to this task order.
- IV.** Attendance at Design Development (90%) Review conference **IS** required with regard to this task order.
- V.** Attendance at Construction Document (100%) submittal conference **IS NOT** required with regard to this task order. Teleconferencing can be utilized for this meeting.
- VI.** Attendance at Construction Pre-Bid conference **IS** required with regard to this task order.

B13 STRUCTURAL ECONOMIC ANALYSIS:

- I. A structural economic analysis **IS NOT** required with regard to this task order.

B14 STRUCTURAL/SEISMIC ANALYSIS:

- I. A structural/seismic analysis **IS** required with regard to this project.

B15 CONSTRUCTION PERIOD SERVICES:

- I. A/E firm shall provide such construction period services to allow the VA to monitor the contract on a daily basis with adequate support from the A/E firm. The items of support shall include, but not be limited to:
1. Review of all technical submittals, shop drawings and other technical correspondence. All submittals and shop drawings shall be reviewed within two weeks of receipt.
 2. During the Construction Phase, the A/E shall attend regular meetings with the Construction Contractor and the VA project team. The Construction Contractor shall prepare agendas and meeting minutes. Construction meeting agendas shall include construction progress, planning, and status of Submittals and RFIs, both outstanding and anticipated.
 3. Provide advice and consultation on RFIs raised by the VA or the contractor on technical matters. Provide additional design and sketches as necessary should issues arise with the constructability of the initial design.
 4. Provide cost estimates as necessary to assist in negotiating any changes to the contract.
 5. A/E firm shall provide adequate representation at all pre-bid and pre-construction conferences to answer questions from contractors and inform them of technical items which should be brought to their attention for a complete job.
- II. VPIH oversight services estimated at **zero (0)** days **ARE NOT** required with regards to this task order.

B16 SITE VISITS:

- I. A/E firm shall provide representatives to make periodic site visits during the construction to provide additional support to the VA in managing the contract.
- II. Site visits during the design period services as outlined in the original contract **ARE** required with regard to this task order. Site visits shall be at a fixed cost per site visit.
1. Initial Site Investigation Visit: Provide **two (2)**, scheduled through the Contracting Officer Representative.
 2. 30% Design Review Visit: Provide **one (1)**, scheduled through the Contracting Officer Representative.

3. 60% Design Review Visit: Provide **one (1)**, scheduled through the Contracting Officer Representative.
4. 90% Design Review Visit: Provide **one (1)**, scheduled through the Contracting Officer Representative.

III. Site visits during the Construction Period Services (CPS) as outlined in the original contract **ARE** required with regard to this task order. Site visits shall be at a fixed cost per site visit.

1. Pre-Bid Visit: Provide **one (1)**, scheduled through the Contracting Officer.
2. General Site Visits during construction: Provide **two (2)**.
3. Final inspections, at completion of project and/or at completion of any phase where the VA must accept work done by the contractor. A final punch list shall be conducted by all design trades. The cumulative findings shall be combined into one final list and provided to the VA for issuance to the contractor: Provide **one (1)**.

All site visits shall be scheduled in advance with the VA COR and CO. Total number of site visits anticipated is **nine (9)**.

IV. Formal documentation will be required for all site visits. Final inspection reports will be prepared by the A/E firm and submitted to the VA for issuance to the contractor. Record of site visits will be in accordance with Special Provisions of the Contract. Correction of A/E's Errors and Omissions, including required site visits to ascertain remedial action, are not part of CPS, and shall be handled under FAR clause 52.236-23.